

AGORUT-D

Trypsin 48 mg + Bromelain 90 mg + Rutoside Trihydrate 100 mg + Diclofenac Sodium 50 mg
Enteric Coated Tablets

1. Pharmacotherapeutic Group

Combination anti-inflammatory and anti-edematous preparation containing:

- Proteolytic enzymes: Trypsin and Bromelain
- Flavonoid antioxidant and capillary stabilizer: Rutoside Trihydrate
- Non-steroidal anti-inflammatory drug (NSAID): Diclofenac Sodium

This combination is classified as an enzyme-based anti-inflammatory and analgesic formulation used in the management of inflammatory conditions associated with edema, pain, and soft tissue injury.

2. Mechanism of Action

Trypsin

Trypsin is a proteolytic enzyme that facilitates the breakdown of inflammatory protein complexes and necrotic tissue debris. It assists in improving microcirculation at the site of injury and promotes tissue remodeling and wound healing.

Bromelain

Bromelain, a proteolytic enzyme derived from pineapple stem, exhibits anti-inflammatory, anti-edematous, and fibrinolytic activity. It reduces vascular permeability and modulates inflammatory mediators including bradykinin and prostaglandins, thereby decreasing swelling and pain.

Rutoside Trihydrate

Rutoside (Rutin) is a bioflavonoid with antioxidant and capillary-protective properties. It stabilizes vascular endothelium, reduces capillary fragility, and helps limit exudation and edema formation in inflamed tissues.

Diclofenac Sodium

Diclofenac is a non-steroidal anti-inflammatory drug (NSAID) that inhibits cyclooxygenase (COX-1 and COX-2) enzymes, leading to reduced prostaglandin synthesis. This results in analgesic, anti-inflammatory, and antipyretic effects.

Synergistic Action

The enzyme-flavonoid combination acts synergistically with Diclofenac by addressing multiple components of inflammation:

- Reduction of inflammatory mediators
- Decrease in tissue edema and microvascular leakage
- Promotion of resorption of hematoma and inflammatory exudates
- Enhancement of tissue repair and recovery

While Diclofenac primarily provides symptomatic relief of pain and inflammation, Trypsin, Bromelain, and Rutoside contribute to reduction of swelling and facilitation of tissue healing.

3. Therapeutic Indications

AGORUT-D is indicated for the management of inflammatory and edematous conditions associated with:

- Musculoskeletal trauma
 - Soft tissue injuries
 - Post-operative inflammation and edema
 - Orthopedic conditions with swelling and pain
 - Sports injuries
 - Dental and maxillofacial surgical procedures
 - Sprains, strains, and contusions
 - Inflammatory conditions associated with impaired tissue recovery
-

4. Posology and Method of Administration

Adults

One tablet twice daily or as directed by the physician.

Method of Administration

- Tablets should be swallowed whole with water.
- Do not crush or chew the enteric-coated tablet.
- Preferably administered after meals to minimize gastrointestinal irritation.

Duration of Therapy

The duration of treatment depends on the severity and clinical response of the condition being treated.

5. Clinical Evidence

Study 1 – Reduction in Post-Traumatic Edema

Clinical evaluations of proteolytic enzyme combinations containing Trypsin and Bromelain have demonstrated significant reduction in soft tissue swelling and edema following trauma and orthopedic surgery. Patients receiving enzyme therapy showed faster reduction in inflammation and improved functional recovery compared with conventional anti-inflammatory therapy alone.

Study 2 – Enhanced Recovery Following Surgery

Studies involving Bromelain-containing anti-inflammatory combinations reported shortened recovery time in post-operative patients, particularly in dental and musculoskeletal procedures. Reduced pain intensity, lower edema scores, and earlier return to normal activity were observed in comparison to control groups receiving standard analgesic therapy alone.

Proteolytic enzyme therapy has also been associated with improved resolution of hematoma and enhanced tissue healing due to modulation of inflammatory pathways and improved microcirculation.

6. Contraindications & Precautions

Contraindications

This product is contraindicated in patients with:

- Hypersensitivity to Diclofenac, Bromelain, Trypsin, Rutoside, or any component of the formulation
- History of NSAID-induced asthma, urticaria, or allergic reactions
- Active peptic ulcer disease or gastrointestinal bleeding
- Severe hepatic impairment
- Severe renal impairment
- Bleeding disorders or coagulation abnormalities
- Concurrent anticoagulant therapy requiring close monitoring

Precautions

Use with caution in:

- Elderly patients
- Patients with mild to moderate renal or hepatic dysfunction
- Patients with history of gastrointestinal ulceration
- Individuals at increased risk of bleeding
- Patients receiving antiplatelet or anticoagulant medications
- Patients with hypertension or cardiovascular disease

Long-term NSAID use may increase the risk of gastrointestinal, renal, and cardiovascular adverse effects.

Pregnancy and Lactation

Use during pregnancy and lactation should be avoided unless clearly indicated and prescribed by a physician. Diclofenac is generally not recommended during the third trimester of pregnancy.

7. Undesirable Effects

The following adverse effects may occur during therapy:

Gastrointestinal Disorders

- Nausea
- Dyspepsia
- Abdominal discomfort
- Gastritis
- Diarrhea
- Peptic ulceration (rare)

Nervous System Disorders

- Headache
- Dizziness

Hypersensitivity Reactions

- Skin rash
- Pruritus
- Urticaria
- Allergic reactions

Hepatic and Renal Effects

- Elevated liver enzymes
- Fluid retention
- Renal function impairment in susceptible patients

Hematologic Effects

- Increased bleeding tendency (rare)

Patients should be advised to discontinue therapy and seek medical attention if symptoms of gastrointestinal bleeding, severe allergic reactions, or significant adverse events occur.

8. Storage

Store below 25°C in a dry place, protected from moisture and direct sunlight.
Keep out of reach of children.

9. Dosage Form

Enteric Coated Tablets.